

Software Engineering for CS

Week 6

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Switch off mobile phones
during lectures,
or
put them into silent mode

Process ?

Why we define processes?

Consistent,

Human dependency (Minimize)

Burger, Variations in taste

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Process Support

One way to minimize errors in the requirements engineering is to use process models and to use CASE tools

The most mature CASE tools support of well-understood activities such as programming and testing and the use of structured methods

Support for requirements engineering is still limited because of the informality and the variability of the process

CASE Tools for RE

Modeling and validation tools support the development of system models which can be used to specify the system and the checking of these models for completeness and consistency

Management tools help manage a database of requirements and support the management of changes to these requirements

Process Improvement

Process improvement is concerned with modifying processes in order to meet some improvement objectives

Improvement objectives

- Quality improvement
- Schedule reduction
- Resource reduction

Planning Process Improvement

Some important questions arise:

What are the problems with current processes?

What are the improvement goals?

How can process improvement be introduced to achieve these goals?

How should process improvements be controlled and managed?

RE Process Problems

Lack of stakeholder involvement

Business needs not considered

Lack of requirements management

Lack of defined responsibilities

Stakeholder communication problems

Over-long schedules and poor quality requirements documents

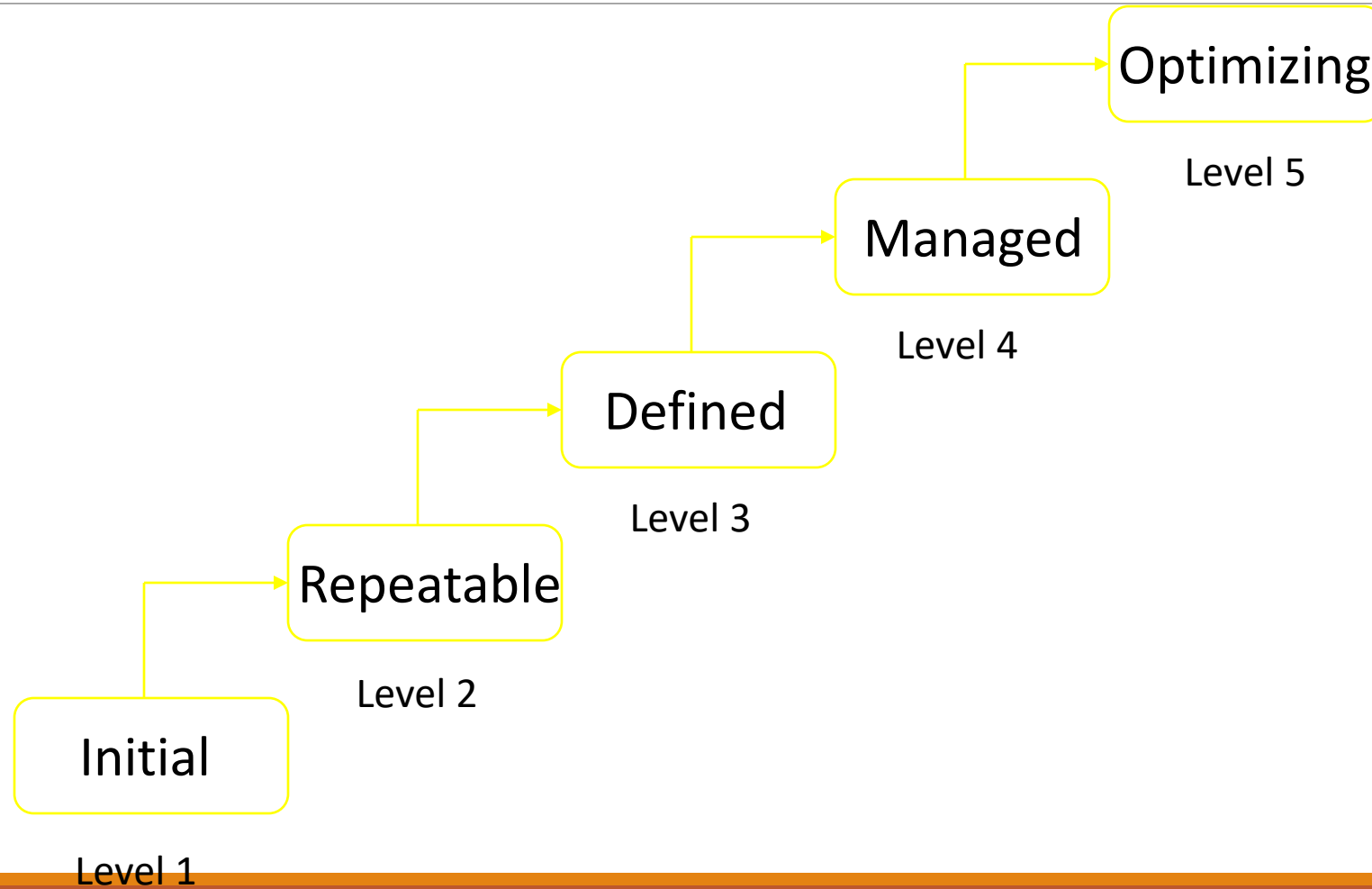
Process Maturity

Process maturity can be thought of as the extent that an organization has defined its processes, actively controls these processes and provides systematic human and computer-based support for them

The SEI's Capability Maturity Model is a framework for assessing software process maturity in development organizations

Capability Maturity Model

Initial – Every organization



CMM Level 1: Initial

Organizations have an undisciplined process and it is left to individuals that how to manage the process and which development techniques to use

CMM Level 2: Repeatable

Organizations have basic cost and schedule management procedures in place. They are likely to be able to make consistent budget and schedule predictions for projects in the same application area

CMM Level 3: Defined

The software process for both management and engineering activities is documented, standardized and integrated into a standard software process for the organization

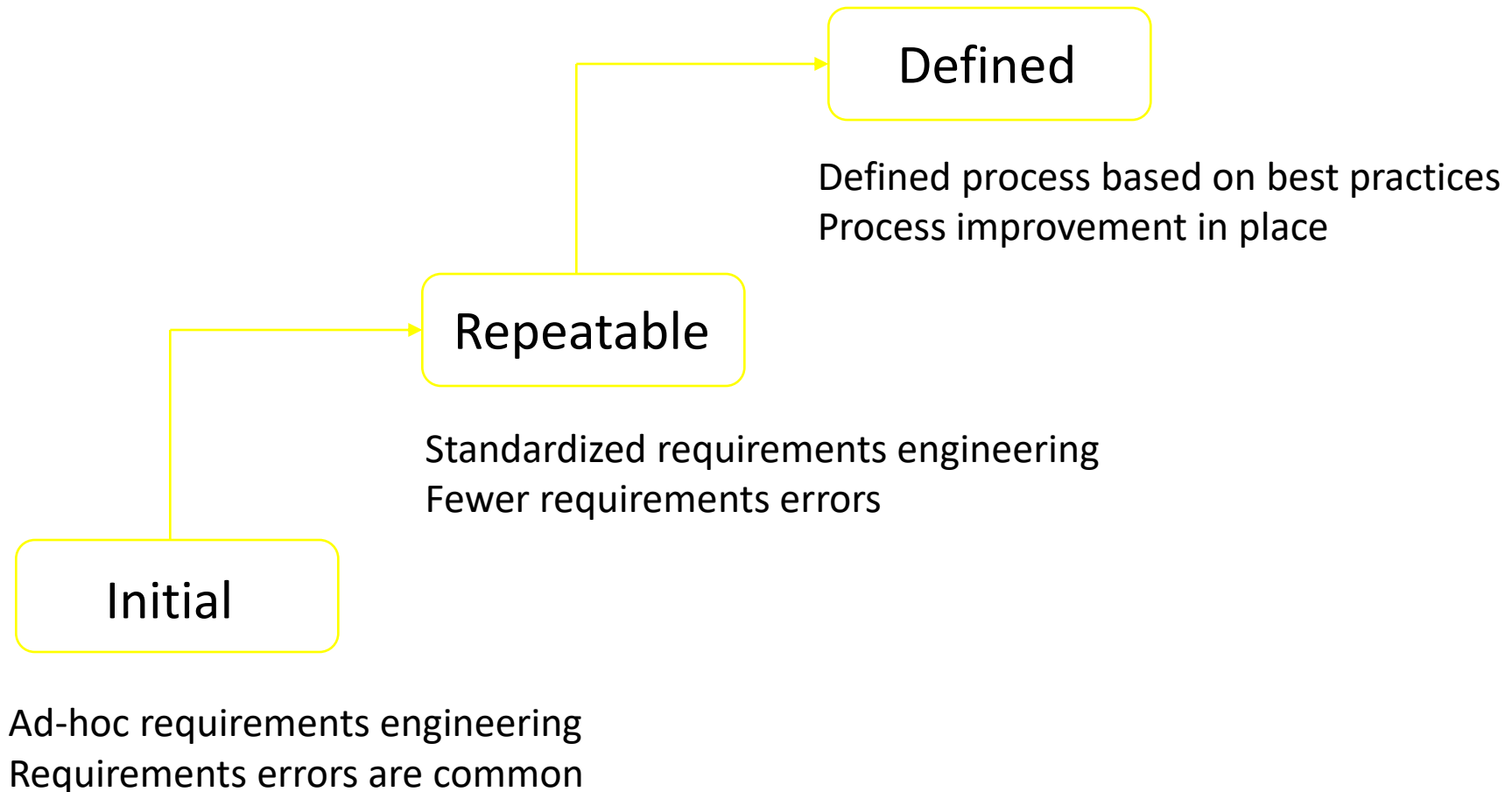
CMM Level 4: Managed

Detailed measurements of both process and product quality are collected and used to control the process

CMM Level 5: Optimizing

The organization has a continuous process improvement strategy, based on objective measurements, in place

RE Process Maturity Model



Initial RE Process Maturity Level

There is no defined RE process.

It suffer from requirements problems such as requirements volatility, unsatisfied stakeholders and high rework costs.

It is dependent on individual skills and experience

Repeatable RE Process Maturity Level

Defined standards for requirements documents, policies and procedures for requirements management

Defined RE Process Maturity Level

Defined RE process based on good practices and techniques. Active process improvement process is in place

Best Practices for RE Process Improvement

RE processes can be improved by the systematic introduction of best requirements engineering practices

Each improvement cycle identifies best practice guidelines and works to introduce them in an organization

Requirements Engineering Costs

About fifteen percent (15%) of system development costs

However, if the requirements engineering process is not executed properly, this cost can increase substantially

Thanks

